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Simplified Direct Displacement Design of Six-Story Woodframe Building and Pretest Seismic Performance Assessment

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Abstract: This paper presents a simplified direct displacement design (DDD) procedure which was used to design the shear walls for a six-story woodframe structure. The building was tested in the final phase of a Network for Earthquake Engineering Simulation (NEES) project. Specifically, *NEESWood Capstone Building* was designed to meet four performance expectations: damage limitation, life safety, far-field collapse prevention (CP), and near-fault CP. The performance expectations are defined in terms of combinations of interstory drift limits and prescribed seismic hazard levels associated with predefined nonexceedance probabilities. To verify that design requirements were met, a series of nonlinear time-history analyses (NLTHAs) was performed using suits of both far-field and near-fault ground motion records. The distributions of interstory drifts obtained from the NLTHA confirm that the Capstone Building designed using DDD meets all four target performance expectations, thereby validating the DDD procedure. Additionally, collapse analysis in accordance with the recently proposed Applied Technology Council project 63 (ATC-63) methodology was performed. The results of incremental dynamic analyses confirmed that the Capstone Building designed using the DDD procedure has adequate capacity margin against collapse, as dictated by the ATC-63 methodology.

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Introduction

In the United States, multistory residential and commercial structures such as multifamily apartments, condominiums, and hotels/motels are often light-frame wood (also known as *woodframe*) construction. For multistory construction, if woodframe is selected over other structural systems, it is typically a result of speed of construction and low construction cost (Cheung 2008). Although woodframe construction provides an economical alternative for multistory buildings, the current U.S. building codes make it difficult to exceed five stories [International Code Council (ICC) 2006] in general and even four stories in some jurisdictions. The height limitation reflects the lack of knowledge of the dynamic response of taller wood buildings under lateral loadings (e.g., wind and earthquake loads), as well as fire safety considerations and other local district land use regulations. Such height restrictions have limited the use of wood for multistory construction in the United States. Nevertheless, many other industrialized countries permit the construction of taller wood buildings (i.e., more than five stories). For example, New Zealand does not have building height restrictions for wood construction. Canada and England have recently revised their building codes to allow the construction of wood buildings of up to six and eight stories, respectively (Craig 2008). In the United States, the timber engineering design and research communities are in the process of developing new design guidelines and procedures that will enable building taller woodframe structures with confidence, including those in seismic regions such as the Pacific Northwest where wood has a strong industry hold. One such effort is the NEESWood project which focuses on the development of a performance-based seismic design (PBSD) procedure for midrise woodframe construction in regions of moderate to high seismicity (van de Lindt et al. 2006).

As part of the NEESWood project, a series of full-scale seismic tests of a two-story Benchmark Woodframe Building was conducted at the Univ. at Buffalo Network for Earthquake Engineering Simulation (NEES) site (Christovasilis et al. 2007). The Benchmark Building was designed in accordance with the *uniform building code* [International Conference of Building Officials (ICBO) 1988]. The test building was representative of a

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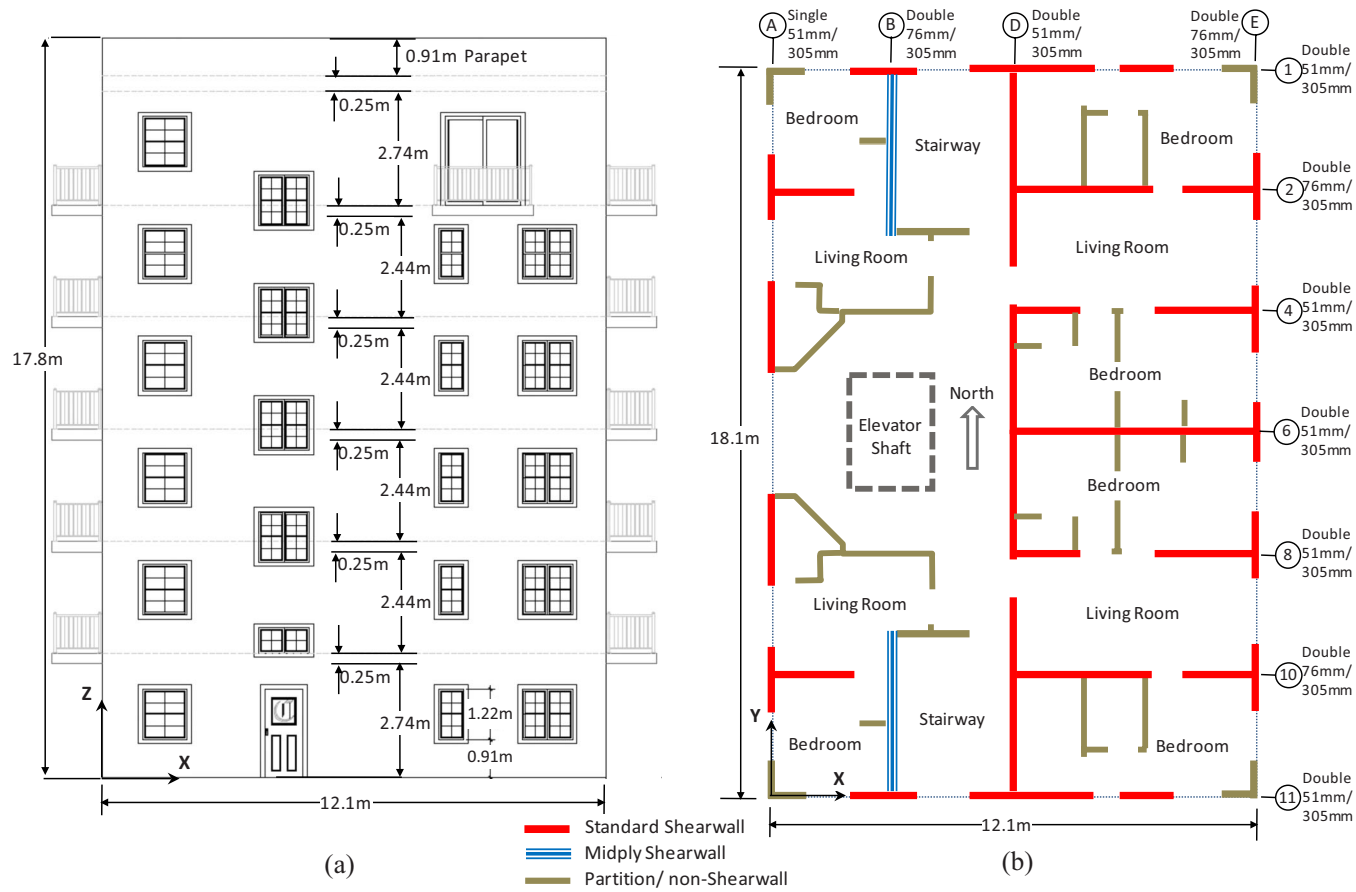

Fig. 1. Architectural layout of the six-story NEESWood Capstone Building: (a) south elevation view; (b) first-floor plan view

typical townhouse structure built in the 1980s and located in the Western United States. The test data collected in the Benchmark Building test included (1) force and deformation measurements of the shear walls and nonload bearing walls; (2) tension force and uplift measurements of the anchor bolts and hold-downs; (3) sill plate slippage; and (4) absolute acceleration measurements. In addition, damages to the structural and nonstructural components, such as the gypsum wall boards (GWBs) and exterior stucco, were visually inspected and documented at the end of each stage of testing. The benchmark test results and findings were used to develop numerical tools and validate a preliminary version of a new direct displacement design (DDD) procedure for PBSD of multistory woodframe buildings (Pang and Rosowsky 2009).

The main objective of the DDD procedure is to limit the building interstory drifts at different levels of seismic hazards. This paper presents the formulation of the new DDD procedure and the application of the proposed procedure in the shear wall design of a six-story woodframe building, which was tested at full scale in the final phase of the NEESWood project. The test building is referred herein as the NEESWood Capstone Building. While not required as part of the DDD procedure, a series of nonlinear time-history analyses (NLTHAs) and incremental dynamic analysis (IDA) (Vamvatsikos and Cornell 2002) was also performed to verify that the as-designed NEESWood Capstone Building meets the target drift limits. The NLTHA and IDA results (presented in later sections) confirmed that the as-designed building meets and exceeds the design requirements.

Description of the Six-Story NEESWood Capstone Building

The architectural layout (Fig. 1) and building design parameters (e.g., the location of bearing walls) determined based on the 2006 *International Building Code* [International Code Council (ICC) 2006] served as the starting point for the displacement-based seismic design of the six-story NEESWood Capstone Building. The plan dimensions of the building are approximately 18.1 m (59.5 ft) in the longitudinal direction and 12.1 m (39.8 ft) in the transverse direction. The height of the building from the base to the top of the roof parapet is approximately 17.5 m (57.5 ft), with a story clear height of 2.74 m (9 ft) for the first story and a story clear height of 2.44 m (8 ft) for second to sixth stories. The thickness of the floor system is approximately 25.4 cm (10 in.) and the roof diaphragm thickness is 38.1 cm (15 in.). The total living space of the test building is approximately 1,350 m^2 (14,500 ft^2). There are 23 living units with four apartment units on each floor except for the sixth floor which contains a large luxury penthouse and two regular apartment units. The total seismic weight of the as-designed building was estimated to be 2,734 kN (615 kip). A series of full-scale shake table tests of the NEESWood Capstone Building was conducted on the E-defense (Miki City) shake table in Japan in July 2009.

Performance Expectations

The six-story Capstone Building was designed to meet the four performance requirements listed in Table 1. Each performance

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Table 1. Performance Expectations for NEESWood Capstone Building

Performance level	Seismic hazard	Performance expectations: Interstory drift limit (%)	Performance expectations: NE probability (%)
Level 1	50%/50 yr	1	50
Level 2	10%/50 yr	2	50
Level 3	2%/50 yr	4	80
Level 4	Near fault	7	50

requirement is specified by a probability of nonexceedance (NE) of an interstory drift limit at a specified level of seismic hazard. The performance requirement is given by the following expression:

$$P_{\mathrm{NE}}(\theta\langle\theta_{\mathrm{lim}}|H) \geq \mathrm{NE}_t \tag{1}$$

where θ and θ_{lim}=interstory drift and target drift limit, respectively. The term $P_{\mathrm{NE}}(\cdot)$=NE probability of the interstory drift at a prescribed hazard level (seismic intensity, H) and NE_t=target/design NE probability. ASCE/SEI-41, *Seismic Rehabilitation of Existing Buildings* [American Society of Civil Engineers (ASCE) 2006], provides guidelines for design and retrofit of structures by specifying three performance levels, namely, immediate occupancy (IO), life safety (LS), and collapse prevention (CP). The IO, LS, and CP definitions correspond to the performance expectations for Levels 1–3 (Table 1) and the hazard levels are associated with earthquakes having 50, 10, and 2% exceedance probabilities in 50 years, respectively. The performance levels/expectations selected by the NEESWood project team and used for designing the Capstone Building are based on the ASCE/SEI-41 guidelines with some modifications. According to ASCI/SEI-41, the interstory drift limits for wood shear walls for the IO, LS, and CP limit states are 1, 2, and 3%, respectively. The NE probabilities for the aforementioned drift limits are assumed to be 50% (median) since the NE probabilities are not explicitly defined in ASCE/SEI-41.

Based on observations made during the NEESWood benchmark test, *nonstructural damage* such as cracking of stucco and GWB occurred at interstory drifts between 0.5 and 1%, and possible *life-safety* related failures such as total splitting of sill plates, buckling of GWB at door/window openings, and separation of GWB from the ceiling were reported at drifts greater than 2% (Christovasilis et al. 2007). Hence, the 1 and 2% drift limits for the IO and LS limit states, respectively, were adopted for Levels 1 and 2 performance expectations without any modifications. It should be noted that while a 1% drift limit with a 50% NE probability was considered to be an "acceptable" drift limit in terms of limiting financial loss, a lower drift limit (e.g., 0.5%) combined with a higher NE probability (e.g., 80%) may be specified in the proposed DDD approach if it is determined that a more stringent damage limitation limit state should be considered.

At Level 3 (2%/50-yr hazard), a drift limit of 4% combined with an 80% NE probability was used as the design performance expectation. The 4% drift limit was based on the benchmark test results for a ground motion representative of 2%/50-yr hazard level where a maximum interstory drift of 3.5% was recorded under a ground motion representative of 2%/50-yr hazard level. At 3.5% drift, the test structure retained about 75% of its lateral initial stiffness and did not exhibit any visible sign of incipient collapse. Hence, the 4% drift limit was selected for Level 3. In the proposed NEESWood performance expectations, buildings located near-fault lines are required to meet the Level 4 performance requirement, namely, a 7% drift limit with a 50% NE probability, when subjected to a suite of near-fault ground motions with strong velocity pulses. The 7% drift limit was based on the collapse analysis of woodframe buildings (Christovasilis et al. 2009) using IDA (Vamvatsikos and Cornell 2002) and has been used in the ATC-63 project to evaluate the collapse probability of wood buildings [Applied Technology Council (ATC) 2008].

Table 2. Design Spectral Acceleration Values for 5% Damping

Hazard level	Intensity (% of DBE)	Exceedance probability	Spectral acceleration: Short-period S_{XS} (g)	Spectral acceleration: 1 s S_{X1} (g)
Short return period earthquake	44	50%/50 yr	0.44	0.26
Design basis earthquake (DBE)	100	10%/50 yr	1.00	0.60
MCE	150	2%/50 yr	1.50	0.90

Note: $X=M$=maximum credible earthquake; D=design basis earthquake; and S=short return period earthquake.

Design Spectra

The Capstone Building is assumed to be located in Southern California and founded on stiff soil (Site Class D). The 5%-damped horizontal acceleration design spectrum parameters for seismic hazard Levels 1–3, determined in accordance with ASCE/SEI-41 [American Society of Civil Engineers (ASCE) 2006], are listed in Table 2. These far-field response spectra (for sites located >10 km from fault rupture) were used in the simplified DDD procedure to design the Capstone Building. Note that the near-fault response spectrum was not specifically determined or used in the design process. However, a suite of unscaled near-fault ground motions (Krawinkler et al. 2003) was used in the NLTHA to verify the design of the Capstone Building at Level 4.

Standard and Midply Shear Walls

The Capstone Building is constructed with North American style engineered light-frame wood shear walls with tie-down systems to restrain uplift forces caused by the overturning moments. The shear walls are built with nominal 51 mm×152 mm (2 in. ×6 in.) Douglas fir and spruce pine fir studs spaced at 406 mm (16 in.) on-center and 10d common nails [3.76 mm in diameter (0.148 in.)] are used to fasten the 11.9-mm (15/32 in.) thick oriented strand board (OSB) to the framing members. The Capstone Building is built almost entirely using conventional North American style stud wall systems (referred as *standard* walls in this paper), except for an interior wall line parallel to the longitudinal direction in which very high shear capacity is required (see Fig. 1) along which a new system known as *midply* construction is used (Varoglu et al. 2007). The midply wall system consists of standard shear wall components but the sheathing is sandwiched between studs that are rotated 90° with respect to those in standard walls and the sheathing is attached to the wide faces of the studs. The sheathing nails in midply walls are driven through studs at one side of the sheathing panel and into studs on the opposite side of the panel resulting in fasteners working in double shears.

The complete shear wall backbone curve is required in the simplified DDD procedure. Both standard and midply shear walls were modeled using the M-CASHEW program, a Matlab version

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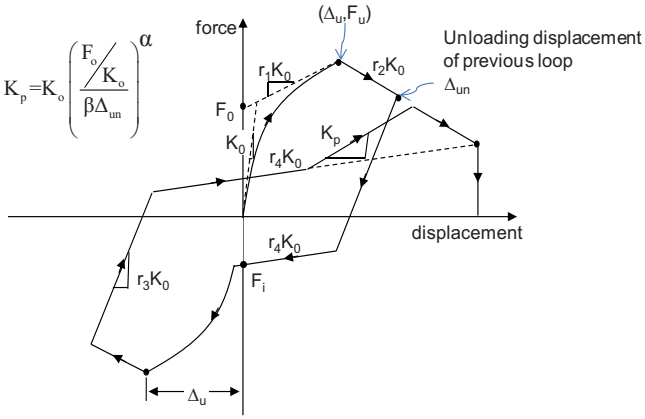

Fig. 2. Modified Stewart hysteretic model

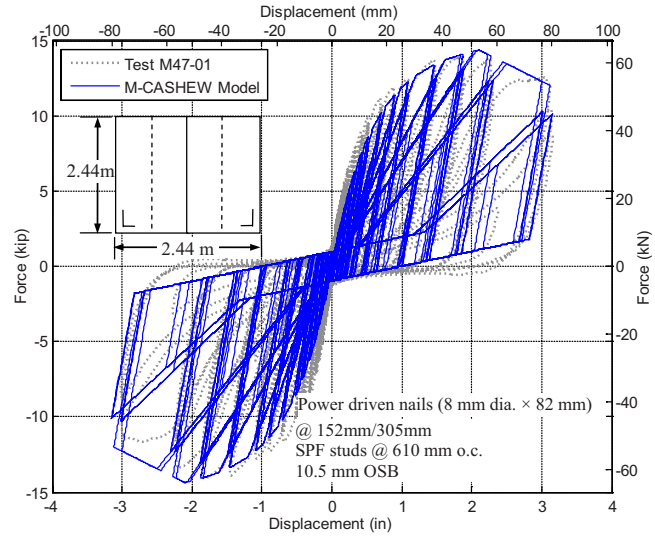

Fig. 3. Model-predicted and test hysteretic loops of midply wall

of the cyclic analysis of wood shear wall (CASHEW) program (Folz and Filiatrault 2001a). The M-CASHEW program can be used to predict the load-displacement response at the top of the wall by modeling the relative movements of the shear wall components (panels and framing members) and the individual load-slip response of nails. In the M-CASHEW model, the nails are modeled using a modified Stewart hysteretic model (Stewart 1987) which includes hysteresis pinching, strength, and stiffness degradation (Fig. 2). The hysteretic parameters for the sheathing nails and dry wall screws are shown in Table 3. Note that the hysteretic parameters for 8d box (2.87 mm in diameter) and 10d common (3.76 mm in diameter) nails were determined by fitting actual cyclic nail test data. The double-shear connector parameters, however, were calibrated by modifying the single-shear nail parameters to match the midply wall test results by Varoglu et al. (2007).

Fig. 3 shows a 2.44 m×2.44 m (8 ft×8 ft) midply shear wall (test M47-01) constructed with nominal 51-mm (2 in.) thick spruce pine fir studs spaced at 610 mm (24 in.) on-center. Sheathing nails were spaced at 102 mm (4 in.) on-center along the panel edges and 203 mm (8 in.) along the interior studs. Power-driven nails, 3 mm (0.118 in.) in diameter and 82 mm (3.23 in.) in length, were used. Since connector data were not available for the actual power-driven nails used to construct the midply test specimen, the parameters of the 8d box nail (having similar diameter) tested in single shear were used to model the test wall. To account for the double-shear effects, the backbone parameters of the nail in single shear were modified by multiplying K_0 and F_0 parameters by 2 and multiplying Δ_u and r_1 parameters by 0.8 (Table 3). This assumption is validated by comparing the hysteretic loops predicted by M-CASHEW with those from the actual midply wall test (Fig. 3). Using the same approach, the parameters for the 10d common nail in double shears were estimated and used to generate the midply backbone parameters used in the displacement-based design of the Capstone Building.

The backbone response of a wood shear wall is given by the following five-parameter equation which consists of a nonlinear logarithmic ascending branch and a linear descending (softening) branch (Fig. 2):

$$F_b(\Delta) = \begin{cases} [1 - e^{-(K_0/F_0)\Delta}(r_1 K_0 \Delta + F_0)] & \text{for } \Delta \le \Delta_u \\ F_u + r_2 K_0(\Delta - \Delta_u) & \text{for } \Delta > \Delta_u \end{cases} \quad (2)$$

Note that Eq. (2) also was used to describe the backbone response of nail and screw connectors. A shear wall database that contains

Table 3. Connector Parameters for Nails in Single and Double Shears

Shear mode	K_0 (kN/mm)	r_1	r_2	r_3	r_4	F_0 (kN)	F_i (kN)	Δ_u (mm)	α	β
8d box nail (2.87-mm diameter) and 9.5-mm OSB										
Single[a]	0.85	0.035	−0.049	1.40	0.015	0.801	0.187	12.19	0.8	1.1
Double	1.71	0.035	−0.0392	1.40	0.015	1.601	0.187	9.75	0.8	1.1
10d common nail (3.76-mm-diameter) and 11.9-mm OSB										
Single[b]	1.55	0.0289	−0.0268	1.04	0.0094	0.979	0.133	8.64	0.73	1.4
Double	3.11	0.0289	−0.0214	1.04	0.0094	1.957	0.133	6.91	0.73	1.4
No. 6 bugle head dry wall screw (3.61 mm diameter×31.75 mm long) and 12.7-mm GWB										
Single[c]	1.75	0.15	−0.03	1.1	0.007	0.445	0.089	5.08	0.8	1.1

[a]Based on the nail test results for nominal 51-mm (2 in.) thick framing member attached to 9.5-mm (3/8 in.) thick OSB using 8d box gun nails.

[b]Based on the cyclic and monotonic nail test results for nominal 51-mm (2 in.) thick hem fir stud attached to 11.9-mm (15/32 in.) thick OSB using 10d common nail (Coyne 2007).

[c]The connector parameters were estimated by matching the M-CASHEW model backbone responses to the actual GWB-only wall test results obtained from the CUREE Task 1.3.1 Test Group 12 (Gatto and Uang 2001) and CUREE Task 1.4.4 Test Group 19 (Pardoen et al. 2003).

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Table 4. Displacement-Based Shear Wall Design Table for Unit Wall Width (per m)

Wall type/panel layer	Edge[a] nail spacing (mm)	K_0 (kN/mm)	F_u (kN)	Backbone force (kN) Drift 0.5%	1.0%	2.0%	3.0%	4.0%
			Wall height=2.74 m					
Standard[b]	51	2.269	31.68	19.42	26.68	31.6	27.36	22.92
	76	1.861	21.37	14.41	18.75	21.22	18.05	14.88
	102	1.586	16.40	11.49	14.53	16.13	13.69	11.24
	152	1.138	11.20	8.12	10.13	11.01	9.44	7.87
Midply[c]	51	2.890	61.53	29.82	46.39	61.52	53.09	44.66
	76	2.514	41.81	23.83	34.75	40.95	35.5	30.05
	102	2.208	31.83	19.76	27.69	30.79	26.77	22.75
	152	1.813	21.70	14.85	19.69	20.93	18.27	15.60
GWB[d]	406	0.743	2.03	1.95	1.85	1.37	0.88	0.39
			Wall height=2.74 m					
Standard[b]	51	2.432	32.2	19.15	26.82	32.05	28.13	23.82
	76	2.176	21.94	14.8	19.17	21.87	18.7	15.52
	102	1.740	16.75	11.64	14.91	16.58	14.18	11.79
	152	1.356	11.41	8.34	10.3	11.27	9.65	8.03
Midply[c]	51	2.971	63.47	28.28	45.33	62.69	55.8	47.52
	76	2.633	42.67	22.94	34.33	41.95	36.58	31.21
	102	2.396	32.26	19.42	27.56	31.5	27.54	23.57
	152	1.988	22.11	14.79	19.87	21.38	18.76	16.14
GWB[d]	406	1.231	2.11	2.04	1.88	1.30	0.73	0.16

[a]Field nail spacing is 305 mm (12 in.).

[b]Standard wall model is built with 11.9-mm-thick OSB connected to framing members by 10d common nails (3.76 mm diameter) in single shear.

[c]Midply wall model is built with 11.9-mm-thick OSB connected to framing members by 10d common nails (3.76 mm diameter) in double shear.

[d]GWB model is built with 12.7-mm-thick GWB connected to framing members by No. 6 bugle head drywall screws (3.61 mm diameter) in single shear.

the backbone parameters for 2.74- (9) and 2.44-m (8 ft) tall standard and midply shear walls with field nail spacing of 305 mm (12 in.) and edge nail spacings of 51, 76, 102, and 152 mm (2, 3, 4, and 6 in.) was generated (Table 4). The shear wall database can also be presented in graphical format (e.g., Fig. 4). Also shown in Table 4 are the backbone parameters for walls sheathed with only 12.7-mm (1/2 in.) thick GWB (i.e., no structural sheathing) connected by 31.75-mm (1.25 in.) long No. 6 bugle head drywall screws at 406 mm (16 in.) on-center. The backbone curve for a wall sheathed with OSB on one side and drywall on the opposite side can be approximated by summing the OSB and GWB backbone curves. This modeling approach has been used by others (White and Ventura 2006; Folz and Filiatrault 2001b; Kim and Rosowsky 2005).

Hysteretic Damping Model for Wood Shear Walls

Hysteretic damping, ξ_{hyst}, in the wood shear wall can be estimated using

$$\xi_{\text{hyst}} = \frac{1}{4\pi}\frac{E_{\text{loop}}}{E_{S0}} = \frac{1}{2\pi}\frac{E_{\text{loop}}}{K_s\Delta_t^2} \tag{3}$$

where E_{loop}=energy dissipated by the actual nonlinear shear wall in one complete cycle and E_{S0}=strain energy of the linear-elastic system at the target displacement, Δ_t, and secant stiffness, K_s, determined at Δ_t. Using the standard and midply shear wall test data obtained from different laboratories (Martin and Skaggs 2003; Varoglu et al. 2007; Pardoen et al. 2003), hysteretic damping values were calculated and plotted in Fig. 5. The results show that the hysteretic damping can be characterized using the secant-to-initial stiffness ratio (K_s/K_0) (Pang et al. 2009)

$$\xi_{\text{hyst}} = 0.32e^{-1.38(K_s/K_o)} \tag{4}$$

Once a target/design wall drift limit has been selected, the secant-to-initial stiffness ratio can be calculated (or interpolated) using the displacement-based shear wall design database (Table 4) and the resulting equivalent hysteretic damping ratio can be computed using Eq. (4). Other studies of hysteretic damping based on the results of cyclic pushover analyses of woodframe structures suggested a hysteretic damping ratio of about 18% of critical when the lateral stiffness of the structure degrades to 33% of its initial stiffness (Filiatrault et al. 2003). At K_s/K_0 of 0.33, the damping model proposed in this study yields an equivalent hysteretic damping of 20% (Fig. 5) which is very close to the value suggested by Filiatrault et al. (2003).

Simplified DDD Procedure

The DDD procedure used to design the shear walls of the six-story NEESWood Capstone Building is a simplified version of the *original* DDD procedure (Pang and Rosowsky 2009). The original DDD procedure was intended to meet specified drift limits with a 50% NE probability (median) and interstory drifts are estimated using a normalized modal analysis which includes contributions from all vibration modes. The main advantages of the new simplified DDD procedure are that (1) it does not require

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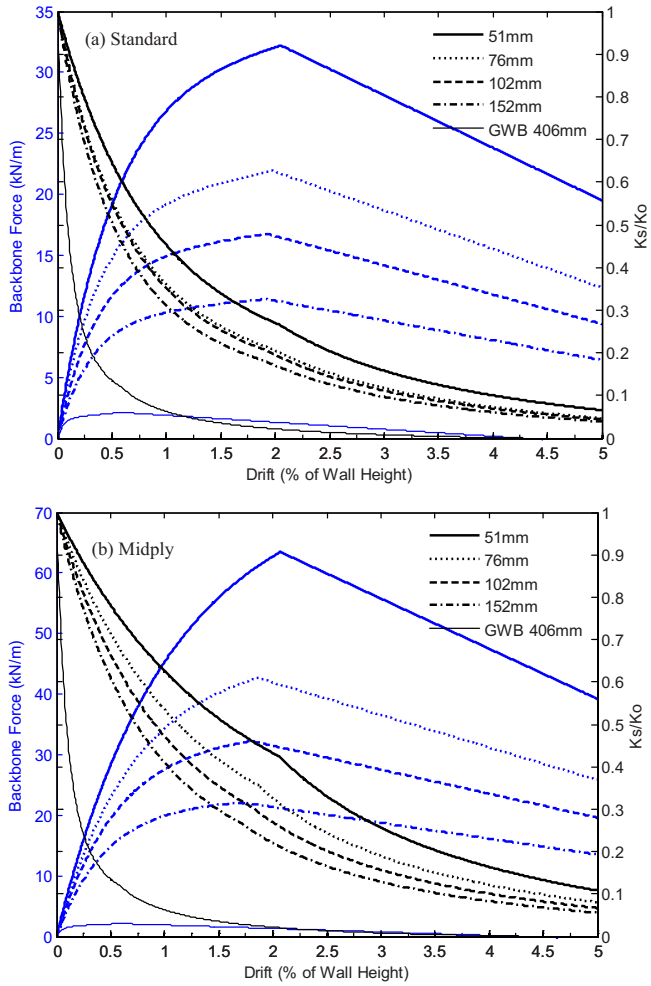

Fig. 4. Shear wall backbone and K_s/K_0 curves for 2.44-m (8 ft) tall (a) standard; (b) midply walls built with 10d common nails and 11.9-mm OSB

modal analysis and thus allows the design to be completed using a spreadsheet and (2) it allows consideration of drift limit NE probabilities other than 50%. It should be noted that the simplified DDD procedure is based on the first-mode approximation and thus it is not applicable to buildings with considerable higher mode effects. Table 5 summarizes the information used to calculate the design forces for Performance Level 3. These design forces were obtained using the simplified DDD procedure, which is described in the following sections.

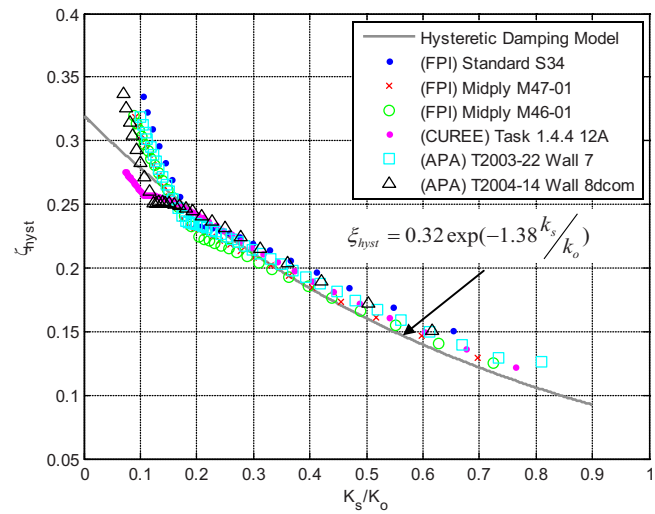

Fig. 5. Equivalent hysteretic damping model

Step 1: Determine the Adjustment Factor for the Specified NE Probability at the Design Drift Limit

The interstory drift limit for seismic hazard Level 3 [maximum considered earthquake (MCE)] is 4% with an 80% NE probability. All other hazard levels were associated with 50% NE probabilities (i.e., median values). The design spectrum specified in both ASCE/SEI-7 [American Society of Civil Engineers (ASCE) 2005] and ASCE/SEI-41 [American Society of Civil Engineers (ASCE) 2006] represents the median demand for the specified hazard level. In order to design for a target NE probability of interstory drift greater than the median, the design spectral value must be adjusted upward to reflect the increase in the design NE probability. The design spectral acceleration adjusted for NE probability, $\vec{S}_X$, is equal to the product of the code-specified spectral acceleration value (median) and the adjustment factor, C_{NE}

$$\vec{S}_X = C_{\mathrm{NE}} S_X \tag{5}$$

The factor C_{NE} is assumed to be log-normally distributed with a median value of 1.0 (assuming that the code-specified median value is unbiased) and a logarithmic standard deviation, β_R, which accounts for the uncertainty of the ground motions, β_{EQ}, as well as the uncertainty associated with the design procedure (i.e., simplified DDD procedure), β_{DS}

Table 5. Summary of DDD Calculations

Story	h_s (m)	h_0 (m)	Δ_{it} (%)	W (kN)	Δ_{it} (mm)	Δ_0 (mm)	$W^*\Delta_0$ (kN mm)	C_v	Δ_v	$C_v^* h_0$ (m)	$W^*\Delta_0^2 \times 10^3$ (kN mm^2)	V_s (kN)	K_s (kN/mm)	F (kN)	$F^* h_0$ (kN m)
1	3.05	3.05	2.13	502	65	65	32,554	0.059	1.000	0.18	2,111	2,185	33.68	129	393.9
2	2.74	5.79	2.13	474	58	123	58,401	0.106	0.941	0.61	7,196	2,055	35.21	232	1,342.6
3	2.74	8.53	2.13	474	58	182	86,064	0.156	0.835	1.33	15,629	1,823	31.24	342	2,915.8
4	2.74	11.28	2.13	474	58	240	113,727	0.207	0.678	2.33	27,290	1,482	25.39	451	5,091.4
5	2.74	14.02	2.13	505	58	298	150,597	0.274	0.472	3.84	44,928	1,030	17.65	598	8,382.0
6	2.74	16.76	2.13	305	58	357	108,965	0.198	0.198	3.32	38,868	433	7.41	433	7,251.4
Σ				2,734	Δ_{eff}=247		550,308	1.000	h_{eff}=11.62		136,022			2,185	25,377.1

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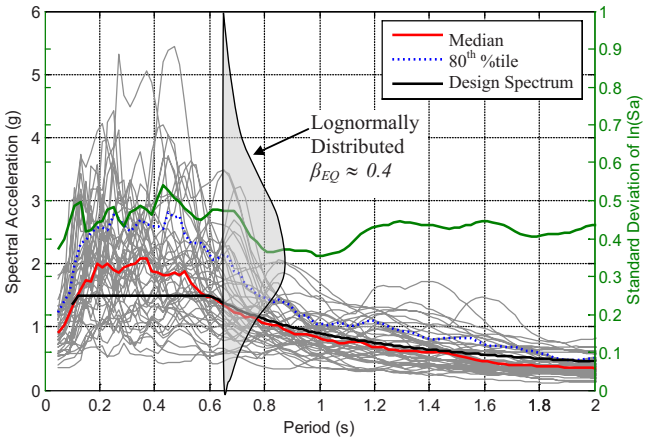

Fig. 6. ATC-63 far-field ground motion ensemble scaled to the Level 3 (MCE) design spectrum

$$\beta_R = \sqrt{\beta_{EQ}^2 + \beta_{DS}^2} \tag{6}$$

Fig. 6 shows the response spectra of the ATC-63 far-field ground motion ensemble scaled to the MCE level (Level 3) and the logarithmic standard deviation of the response spectra, β_{EQ}. The uncertainty due to the ground motion varies from about 0.35 to 0.5. Following the ATC-63 study, a fixed value of 0.4 was assumed for the β_{EQ}. The simplified DDD procedure does not explicitly account for a number of factors that might affect the actual interstory drift response such as torsion, higher mode effects, anchor tie-down system (continuous rod) elongation and compression of the chord members, or flexible diaphragms. The uncertainty introduced into the analysis arising from these assumptions/simplifications, β_{DS}, was assumed to be 0.6 and the total uncertainty β_R, rounded up to the nearest 0.05, was determined to be 0.75 using Eq. (6). This total uncertainty (0.75) is the same as using the ATC-63 quality ratings of "fair," "good," and good for the modeling, test data, and design requirement uncertainties, respectively [Applied Technology Council (ATC) 2008].

The adjustment factor for the NE probability, C_{NE}, can be determined using the inverse of the lognormal cumulative distribution function (CDF) with median value of 1.0 (logarithmic median=0)

$$C_{\mathrm{NE}} = \exp[\Phi^{-1}(\mathrm{NE}_t)\beta_R + \ln(1)] = \exp[\Phi^{-1}(\mathrm{NE}_t)\beta_R] \tag{7}$$

where $\Phi^{-1}(\cdot)$ is the inverse CDF of the standard normal distribution. Using the inverse cumulative lognormal distribution, the C_{NE} factor for the Capstone Building Level 3 design with 80% NE probability was determined to be 1.88.

Step 2: Select a Design Interstory Drift

The proposed NEESWood drift limit for seismic hazard Level 3 is 4%. The design interstory drift adjusted for NE probability was $4\%/C_{\mathrm{NE}}=2.13\%$, an equivalent 50% NE drift limit, θ_{eq50}. Fig. 7 shows the target peak interstory drift curve for seismic hazard Level 3. Note that the median of the new peak interstory drift distribution curve is equal to θ_{eq50}. The equivalent 50% NE interstory drift limit was used in the displacement-based design of the six-story building (Table 5). While a constant interstory drift limit was used throughout the design of the six-story Capstone Building, the procedure allows different interstory drift limits to be assigned to each story if desired.

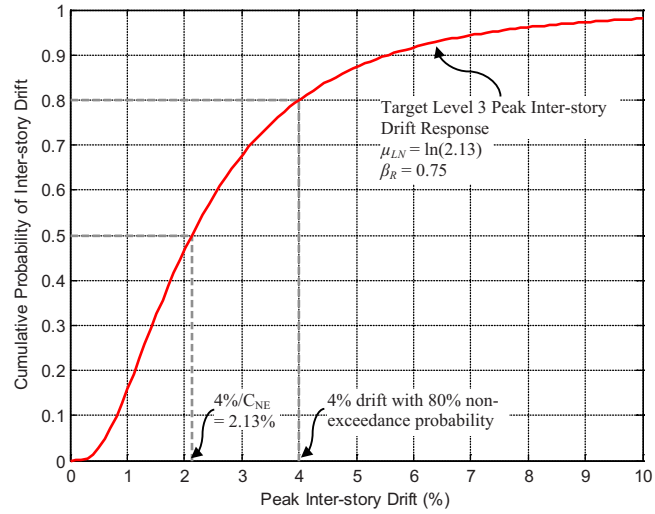

Fig. 7. Target peak interstory drift distribution curve

Step 3: Calculate the Vertical Distribution Factors for Base Shear, C_v

The vertical distribution factors for base shear, C_v, are computed as

$$C_{v_j} = \frac{W_j \Delta_{0j}}{\sum_i W_i \Delta_{0i}} \tag{8}$$

where subscript i=floor number; W=lumped seismic weight of the floor or the roof diaphragm; and Δ_0=target floor displacement relative to the ground (Fig. 8). The seismic weights listed in Table 5 were estimated based on the tributary area of the shear walls (i.e., half of the wall weight was assigned to the floor above and half to the floor below).

Step 4: Calculate the Effective Height, h_{eff}, for the Substitute Structure Modeled as a SDOF System

The effective height is located at the centroid of the assumed lateral force distribution and is calculated as

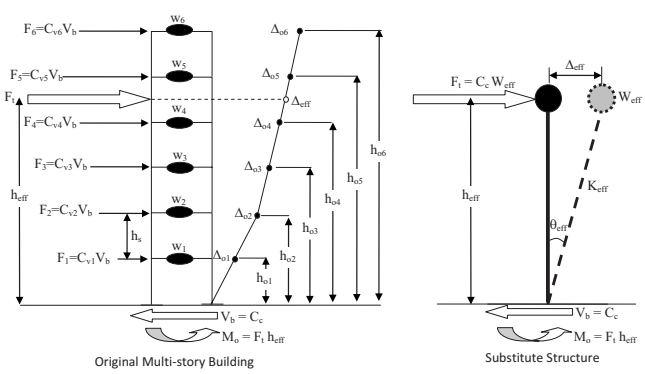

Fig. 8. Example of six-story building and substitute structure for DDD procedure

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$$h_{\text{eff}} = \frac{\sum_i C_{v_i} h_{0i}}{\sum_i C_{v_i} = 1} = \sum_i C_{v_i} h_{0i} \tag{9}$$

where β_{vi}=story shear factor computed as the sum of the vertical distribution factors, c_{vi}, on and above the ith floor and h_0=floor height with respect to the ground. For typical multistory buildings with approximately equal story heights and seismic weights at each story, the ratio of effective-to-roof height generally is about 0.7. The effective height for the six-story Capstone Building was determined to be 11.62 m (Table 5) or 0.69 times the roof height.

Step 5: Use Interpolation to Obtain the Target Displacement at the Effective Height, Δ_{eff}, or Target Drift at Effective Height, θ_{eff}

The effective height (11.62 m) for the NEESWood Capstone Building is located between Levels 4 and 5 (Table 5). Using interpolation, the effective displacement with respect to the ground level is 247 mm (9.73 in.).

Step 6: Calculate the Effective Seismic Weight, W_{eff}, of the Substitute Structure

The effective seismic weight, W_{eff}, of the substitute structure is calculated as

$$W_{\text{eff}} = \frac{\left(\sum_i W_i \Delta_{0i}\right)^2}{\sum_i W_i \Delta_{0i}^2} \tag{10}$$

The $\Sigma_i W_i \Delta_{0i}$ and $\Sigma_i W_i \Delta_{0i}^2$ terms are shown in last row of Table 5. The effective seismic weight for the six-story NEESWood Capstone Building is $(550,308)^2/(136,022\times 10^3)=2,226$ kN (500.5 kip). For most midrise buildings of regular plan, the effective seismic weight usually is about 80% of the total seismic weight. For the Capstone Building, the effective seismic weight is 81% of its total weight.

Step 7: Determine the Damping Reduction Factor, B_ξ

The damping reduction factor, B_ξ, is determined per ASCE/SEI-41 [American Society of Civil Engineers (ASCE) 2006], Sec. 1.6.1.5, as

$$B_\xi = \frac{4}{5.5 - \ln(100\xi_{\text{eff}})} \tag{11}$$

where ξ_{eff}=effective viscous damping as a fraction of the critical damping, computed as the sum of the hysteretic damping, ξ_{hyst} [see Eq. (4)], and the intrinsic damping, ξ_{int}

$$\xi_{\text{eff}} = \xi_{\text{int}} + \xi_{\text{hyst}} \tag{12}$$

In the design of the six-story NEESWood Capstone Building, 5% intrinsic damping was assumed. At the adjusted design drift limit of 2.13% and assuming most walls are built with a 51-mm (2 in.) or 76-mm (3 in.) perimeter nailing, the K_s/K_0 ratio is about 0.30 (Fig. 4). Substituting K_s/K_0 of 0.30 into the equivalent hysteretic damping equation [Eq. (4)] gives an estimated hysteretic damping of 0.21. The total equivalent viscous damping, including the intrinsic damping, therefore is 0.26. Using Eq. (11), the damping reduction factor therefore is 1.71.

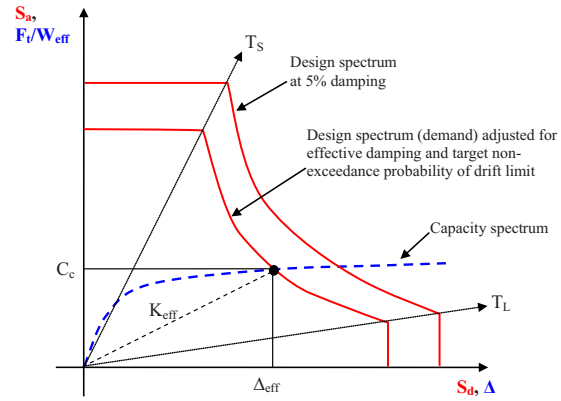

Fig. 9. Determination of the design base shear coefficient using capacity spectrum approach

Step 8: Determine the Design Base Shear Coefficient, C_c

The design base shear coefficient, C_c, is determined using the capacity spectrum approach as

$$C_c = \min\begin{cases} \dfrac{C_{\text{NE}} S_{XS}}{B_\xi} \\ \dfrac{g}{4\pi^2 \Delta_{\text{eff}}}\left(\dfrac{C_{\text{NE}} S_{X1}}{B_\xi}\right)^2 \end{cases} \tag{13}$$

Eq. (13) is the solution for the intersection between the demand and the capacity spectra (Shama and Mander 2003) (Fig. 9). For seismic hazard Level 3, the spectral design values for short-period, S_{MS}, and 1-s period, S_{M1}, are 0.9 and 1.5 g, respectively (Table 2). The first term in Eq. (13) is for a structure having a secant period (at the design displacement, Δ_{eff}) less than or equal to the short period, T_s, defined in Sec. 11.4 of ASCE/SEI-7 [American Society of Civil Engineers (ASCE) 2005]. For most midrise buildings, where the secant periods are generally greater than T_s but less than T_L, the second term usually governs the design. The long-period transition period, T_L, can be obtained from ASCE/SEI-7 [American Society of Civil Engineers (ASCE) 2005]. Using Eq. (9), the base shear coefficient for seismic hazard Level 3 therefore is 0.981.

Step 9: Calculate Design Forces

Once the base shear coefficient is obtained, the base shear, the lateral forces, the story shears, the overturning moments, and the required story secant stiffnesses are calculated as follow:

Base shear, V_b

$$V_b = C_c W_{\text{eff}} \tag{14}$$

equivalent static lateral forces, F_i

$$F_i = C_{v_i} C_c W_{\text{eff}} = C_{v_i} V_b \tag{15}$$

story shears, V_{s_i}

$$V_{s_i} = \beta_{v_i} V_b \tag{16}$$

overturning moment, M_{0_i}

$$M_{0_i} = \sum_{j=i}^{N_s} F_j (h_j - h_i) \tag{17}$$

where N_s=total number of stories (i.e., six for the Capstone Building), and effective secant stiffness (SDOF), K_{eff}

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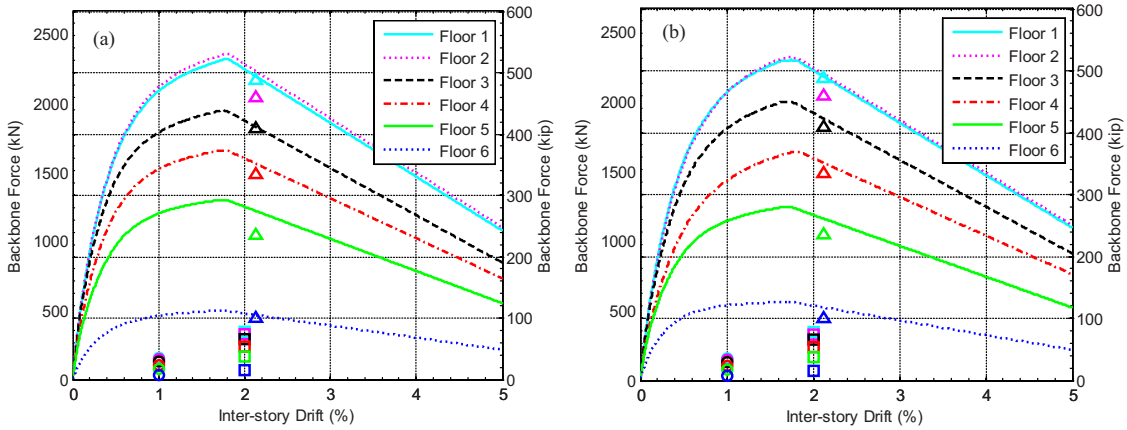

Fig. 10. Design points for seismic hazard Level 3 and interstory backbone curves in (a) transverse; (b) longitudinal directions

$$K_{\text{eff}} = \frac{C_c W_{\text{eff}}}{\theta_{\text{eff}} h_{\text{eff}}} = \frac{C_c W_{\text{eff}}}{\Delta_{\text{eff}}} \tag{18}$$

The required secant stiffness for each story

$$K_{s_i} = \frac{V_{S_i}}{\Delta_{\text{it}_i}} \tag{19}$$

From Table 5, the design base shear and overturning moment are approximately 2,185 kN (491 kip) and 25,377 kN m (18,718 kip ft), respectively. The required effective secant stiffness of the building at the target drift limit, computed using Eq. (18), is 8.84 kN/mm (50.47 kip/in.). The effective secant period, computed as $2\pi/\sqrt{(gK_{\text{eff}}/W_{\text{eff}})}$, therefore is 1.01 s. Recall that the secant-to-initial stiffness ratio of 0.30 was assumed when determining the hysteretic damping, the minimum initial design stiffness therefore is $K_{\text{eff}}/0.30 = 29.46$ kN/mm (168.23 kip/in.), and the associated initial period is 0.55 s.

Step 10: Select Shear Walls to Meet the Design Story Shears

The design points or the expected design interstory drift and required story shear pairs (θ_{it} and V_s) are shown in Table 5. Shear wall nailing schedules were selected from the shear wall database (Table 4). Shear wall backbone forces were taken from the "2% drift" column since the adjusted design interstory drift was determined to be 2.13% for seismic hazard Level 3. The design story shears were distributed to wall lines according to their tributary areas. Direct summation of the equivalent stiffness of shear wall segments was used to generate the story backbone curves. Note that this assumes no torsion and that all shear walls at the same floor level experience the same drift. The nailing patterns for the shear walls for each floor were determined such that the story backbone curve was above the design points associated with that floor (see Fig. 10). In the first story, most of the standard shear walls are sheathed with two layers of OSB (one layer on each side of the wall) attached using nails with either a 51-mm (2-in.) or 76-mm (3-in.) edge spacing (Fig. 1). At wall Line B (parallel to the longitudinal direction) in the first story, double-layer midply shear walls with 76-mm (3-in.) edge nail spacing were used. The complete shear wall nail schedules for Stories 1–6 are given by Pang et al. (2009).

Although not required as part of the DDD procedure, a series of NLTHAs was performed using suits of far-field and near-fault ground motion records to verify that design requirements were met. The as-designed six-story Capstone Building was analyzed using both two-dimensional (2D) and three-dimensional (3D) numerical models and the NLTHA results are discussed in the subsequent sections.

2D NLTHA

In seismic design, engineered light-frame wood buildings are generally considered as shear-dominant systems. Thus, a numerical model for the Capstone Building was constructed using the M-SAWS program (Pang et al. 2009), a Matlab version of the SAWS program (Folz and Filiatrault 2001b), which considers only the pure-shear deformation of the shear walls. In the M-SAWS model, rigid diaphragms with one rotational and two in-plane translational degrees of freedom are assumed for each floor and roof diaphragm. Each shear wall was modeled as a zero-height nonlinear SDOF spring using the modified Stewart hysteretic model. In this paper, the M-SAWS model is referred to as the 2D model. The first three mode shapes and periods of the Capstone Building, based on the 2D model, are shown in Fig. 11. The estimated effective fundamental period is about 0.50 s which corresponds to a primary translational mode shape drift in the Y (longitudinal) direction. The second mode is a pure translational mode in the X (transverse) direction with negligible rotation. Mode 3 is a pure rotational or torsional mode which causes the building to twist around the center of gravity of the floor diaphragms. Since most of the damping is accounted for in the hysteresis model itself, only a very low amount of viscous damping (2% of critical damping) was used in the NLTHA.

3D NLTHA

Although light-frame wood buildings generally are treated as lateral shear-dominant systems in design, they can also be affected by vertical excitation and overturning moment which induces tension forces in the shear wall hold-down system and can cause cumulative elongation of the hold-down rods, especially in buildings exceeding three stories. Pei and van de Lindt (2009) developed a simplified model that is capable of incorporating the effect of overturning and uplift as well as the vertical ground motion excitation in the seismic responses of woodframe structures. The proposed model for wood structures with vertical/uplift effects is quite different from the shear-only model in that the model assigns six degrees of freedom at each story diaphragm and includes the stiffness of the hold-down system and the vertical stiffness of shear walls provided by the vertical framing members.

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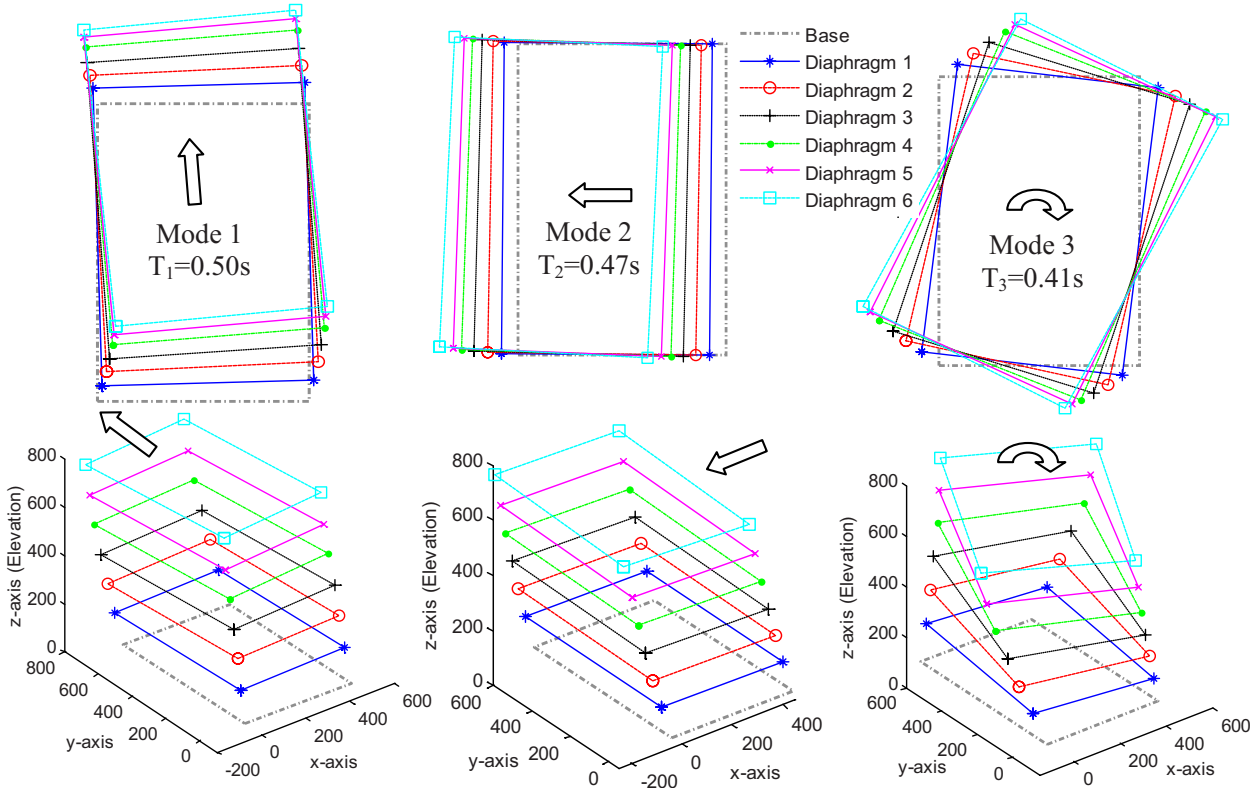

Fig. 11. First three mode shapes of the 2D numerical model

The diaphragm is allowed to move and rotate out of the horizontal plane, adding another dimension to the dynamic analysis to make it three dimensions. The effect of vertical ground motion on the tie-down systems is taken into account through an unsymmetrical linear spring (difference stiffness in tension and compression, representing the stiffness of tie-down rods and compression stud packs), as well as the lateral response resulting from diaphragm rotation. The 3D model has been implemented into the SAPWood program, developed as part of the NEESWood project. In addition to the M-SAWS model, the SAPWood program was used to perform the 3D NLTHA to verify the applicability of both the M-SAWS model and the DDD procedure.

Ground Motions

Two sets of ground motion ensembles were considered in the 2D NLTHA: (1) 22 biaxial ATC-63 far-field ground motions scaled according to the ATC-63 methodology [Applied Technology Council (ATC) 2008] for seismic hazard Levels 1–3 and (2) six biaxial CUREE unscaled near-fault ground motions (Krawinkler et al. 2003; Pang et al. 2009) for seismic hazard Level 4. For hazard Levels 1–3, the median response spectrum of the normalized ground motion ensemble was scaled using a single scaling factor to match the design 5%-damped spectral acceleration at the fundamental period of the building [Applied Technology Council (ATC) 2008]. The scale factors for the 22-biaxial ATC-63 ground motions are given by Pang et al. (2009). The ground motions were scaled and the building was analyzed at each of the three performance levels. The biaxial ground motions were also rotated by 90° and thus, at each performance level, the building was analyzed twice for each of the 22 record pairs for a total of 44 analyses. Similarly, the building was also analyzed using the six pairs of near-fault ground motions rotated at 0° and 90° for seismic hazard Level 4.

Expected Peak Interstory Drift Distributions

The peak interstory drift distributions based on the results from the 3D and 2D NLTHAs are shown in Fig. 12 as solid lines and dotted lines, respectively, and the corresponding NE probabilities at the design drift limits are summarized in Table 6. Including the vertical effect generally results in slightly higher peak interstory drifts than those obtained from the 2D NLTHA (i.e., moving the

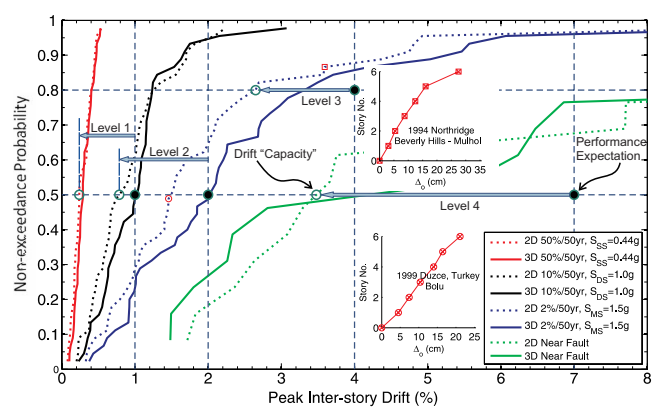

Fig. 12. Peak interstory drift distributions of the NEESWood Capstone Building

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Table 6. Summary of NLTHAs of Six-Story NEESWood Capstone Building Designed using DDD

Seismic hazard		Performance expectation		2D NLTHA			3D NLTHA		
Level	Ground motion	Drift limit (%)	P_{NE} (%)	Median drift (%)	P_{NE} at drift limit (%)	Pass?	Median drift (%)	P_{NE} at drift limit (%)	Pass?
1	50%/50 yr	1	50	0.24	>98	Yes	0.29	>98	Yes
2	10%/50 yr	2	50	0.78	95	Yes	1.02	95	Yes
3	2%/50 yr	4	80	1.49	88	Yes	2.03	86	Yes
4	Near fault	7	50	3.48	68	Yes	4.19	77	Yes

peak interstory drift curves to the right). However, for the six-story woodframe structure designed in this study, the differences in the interstory drifts between the shear-only (2D) model and the 3D model are not felt to be significant. This result is not unexpected because of the aspect ratio (lateral dimension/height approximately equal to 1) of the building that makes the dynamic behavior shear dominant, which is commonly seen in most typical woodframe building floor plans, i.e., multiunit residential structures.

In summary, both the 2D and 3D NLTHAs indicate that the Capstone Building designed using the simplified DDD procedure satisfies all four design objectives. The Capstone Building designed using DDD procedure performs satisfactorily (i.e., meets performance requirements) at all four hazard levels. The median peak drifts at Levels 1 and 2 were considerably lower than the 1 and 2% drift limits, while the median peak drift at Level 3 was 1.49% with an 83% probability of not exceeding the 4% drift limit. At Level 4, the probability of exceeding the 7% drift limit was approximately 32% which satisfied the near-fault ground motion performance requirement. The drift profiles (relative to the ground) of the two selected earthquake records at the MCE level (2%/50 yr) are also shown in Fig. 12. It can be seen that the drift profiles are relatively uniform, which means the seismic demand was distributed evenly among the stories. In other words, the Capstone Building does not have a significant "weak story."

ATC-63 CMR

In addition to considering the four NEESWood performance requirements, monotonic pushover and IDAs (Vamvatsikos and Cornell 2002) were performed to evaluate the collapse margin ratio (CMR) of the test building using the ATC-63 methodology [Applied Technology Council (ATC) 2008]. The ATC-63 methodology was developed for evaluating the collapse risk of structures designed using the current code-specified force-based procedures under MCE ground motions. An evaluation of the CMR using the ATC-63 procedure provides additional perspective on collapse risk of the Capstone Building designed using the DDD procedure. To compute the collapse capacity, IDA was performed using the ATC-63 far-field ground motions. The spectral intensity of the ground motion causing 50% of the analyses/cases to collapse is 2.57 g and the unadjusted CMR is 2.57/1.50=1.71 (Fig. 13). According to the ATC-63 methodology, the raw CMR must be adjusted for the spectral shape before the acceptance criterion can be determined.

The spectral shape factor (SSF) is a function of the seismic design category (SDC), ductility of the structure which is determined through the pushover curve and the upper limit of the code-defined fundamental period of the structure [Applied Technology Council (ATC) 2008]. The Capstone Building is designed for SDC D_{max} (Southern California regions) and the code-defined period, determined per ASCE/SEI-07 Sec. 12.8.2, is 0.57 s. Fig. 14 shows the monotonic pushover curve obtained by applying an inverted triangular lateral load parallel to the transverse direction (x axis) of the test building. The maximum base shear in the transverse direction is 2,734 kN (514.7 kip) and occurs at a roof drift ratio of 1.27% (roof height is 16.76 m). The seismic coefficient at the peak of the pushover curve, V/W, is 0.849. The ultimate drift (1.54%) is defined at the point where the base shear deteriorates to 80% of the maximum value. An idealized elastic-plastic curve is determined by defining the initial stiffness using a

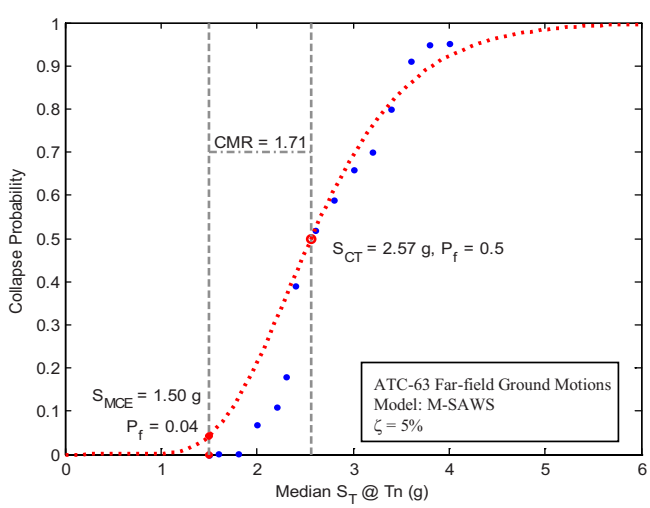

Fig. 13. Collapse fragility curve of the NEESWood Capstone Building

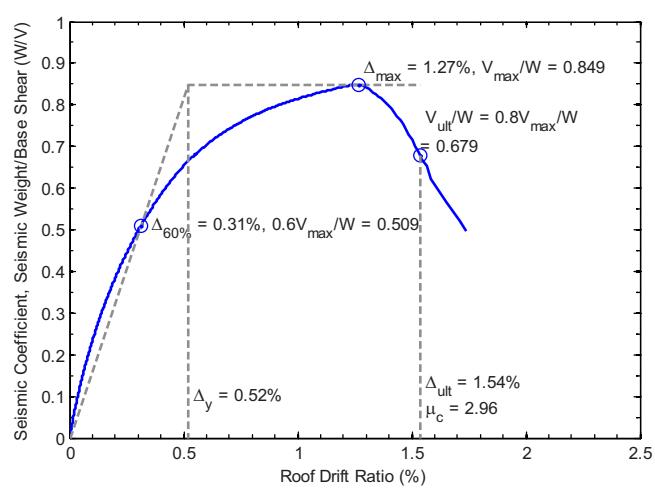

Fig. 14. Monotonic pushover curve (transverse, X-direction) of the NEESWood Capstone Building

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secant-stiffness line that passes through the point where the base shear is at 60% of the maximum. From the elastic-plastic curve, the "yield" drift is 0.52% and the ductility factor, μ_c, is computed as 1.54/0.52=2.96. Using Table B-4 in the ATC-63 90% draft report, the SSF is 1.22 [Applied Technology Council (ATC) 2008]. Therefore, the adjusted CMR (ACMR) is computed as CMR×SSF=2.09. While only the pushover response in the transverse direction is discussed herein, it should be noted that the pushover curve in the longitudinal direction is very similar to that in the transverse direction. This is because the interstory backbone curves of the Capstone Building designed using the DDD procedure are very similar in two horizontal directions (Fig. 10). Therefore, the ACMRs are approximately the same in both directions.

The acceptable value for the ACMR of an individual system (i.e., <20% collapse probability) depends on the uncertainties of the model and the design procedure. Using the same assumptions as the ATC-63 wood building design examples, the uncertainty in ground motion records is 0.40, design requirement uncertainty (B: good) is 0.30, test data quality (B: good) is 0.30, and modeling uncertainty (C: fair) is 0.45. Thus the composite/total uncertainty, β_{tot}, is 0.75 [Table 7-2c in Applied Technology Council (ATC) 2008]. The Capstone Building satisfies the ATC-63 collapse margin requirement since the ACMR of the Capstone Building (2.09) is higher than the acceptable ACMR for individual building with β_{tot} of 0.75, which is 1.88 [determined from Table 7-3 in Applied Technology Council (ATC) 2008]. Based on the adjusted collapse fragility curve, the collapse probability of the Capstone Building at MCE level is approximately 16%.

Summary and Discussion

A simplified DDD procedure for performance-based design of multistory wood buildings is presented. The design procedure can be used to consider drift limit NE probabilities other than 50%. The proposed design procedure is relatively simple and the shear wall design process can be performed using a spreadsheet. The simplified DDD procedure was used to design the shear walls of the six-story NEESWood Capstone Building. To validate the design procedure, two numerical models (2D and 3D models) were constructed and NLTHAs were performed using the ATC-63 far-field ground motions and a set of near-fault ground motions. The results of the NLTHA confirmed that the Capstone Building designed using the simplified DDD procedure satisfies all four design performance requirements. Additionally, the results of the NLTHA show that the seismic demand was distributed evenly among the stories (uniform drift profiles). Finally, the CMR of the Capstone Building under MCE ground motions was determined to be acceptable per the ATC-63 methodology.

In the simplified DDD procedure, an adjustment factor C_{NE} was introduced to design for performance requirements associated with NE probabilities other than the median. While it is possible to determine C_{NE} for each specific building using the procedure outlined in this study, the current procedure for determining C_{NE} requires the engineers to be familiar with fragility analysis and the treatment of uncertainties at the outset. This may be viewed as a disadvantage of the procedure since a rigorous fragility analysis with proper treatment of uncertainties is a time-consuming process which might not be justifiable for typical residential building designs. One possible way to address this drawback is to preanalyze a portfolio of buildings (e.g., the ATC-63 woodframe structure archetypes) and develop design charts or tables for determining C_{NE} for use in the simplified DDD procedure. Then, design charts can be created for selection of the adjustment factor considering different NE probabilities. This would provide a relatively simple procedure for DDD of multistory woodframe buildings in which the engineer is given flexibility in setting NE probabilities associated with the different performance requirements/drift limits.

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